

ZMP-based stability of the mobile manipulator

1. Mechanical model and derivation of the stability measure

Aim. Define a physically interpretable tipping-stability objective for coordinated base, lift, and arm motion. The formulation uses Newton-Euler dynamics and Lie-group kinematics; it does not assume a particular control algorithm.

1. System and mechanical inputs

$$q = (G, s), \quad G \in SE(3), \quad s = (h, \theta, \phi, \rho), \quad v = (\xi_B, \dot{s})$$

$$\dot{G} = G\hat{\xi}_B, \quad \xi_B = (\omega_B, v_B), \quad \dot{s} = v_s$$

$$M(q)\dot{v} + c(q, v) + g_q(q) + d(q, v) = B(q)u_{\text{mech}} + J_c(q)^T \lambda_c$$

G is the full chassis pose; s contains lift height h , seven arm angles, four wheel angles, and passive roller angles. M is inertia, c is the velocity-dependent inertial term, g_q is gravity, d is resistance, B maps actuator effort, and J_c maps ground contact. The mechanical inputs are lift force F_h , seven arm torques and four wheel torques. Scissor-link motion is parameterized by h .

2. Whole-system Newton-Euler balance

$$m = \sum_i m_i, \quad p_C = \frac{1}{m} \sum_i m_i p_{C,i}$$

$$H_C = \sum_i [R_i I_i \omega_i^B + (p_{C,i} - p_C) \times m_i (\dot{p}_{C,i} - \dot{p}_C)]$$

For component i : m_i is mass, $p_{C,i}$ is its world centre of mass, R_i is body-to-world rotation, I_i is inertia about its centre of mass, and ω_i^B is angular velocity in its own body frame. H_C is total angular momentum about the whole-system centre of mass. All terms in H_C are expressed in the world frame.

$$F = m(\ddot{p}_C + g e_3), \quad M_O = \dot{H}_C + p_C \times F$$

F is the total ground force; M_O is its moment about a fixed floor origin; e_3 is the upward unit vector. These balances assume gravity and ground contact are the only external loads. An external tool wrench must be added when manipulating the environment.

3. Zero-moment point on the floor

$$x_Z = -\frac{M_{O,y}}{F_z}, \quad y_Z = \frac{M_{O,x}}{F_z}, \quad F_z > 0$$

$$x_Z = x_C - \frac{z_C m \ddot{x}_C + \dot{H}_{C,y}}{m(g + \ddot{z}_C)}$$

$$y_Z = y_C - \frac{z_C m \ddot{y}_C - \dot{H}_{C,x}}{m(g + \ddot{z}_C)}$$

The floor is horizontal at $z = 0$. Positive F_z is required. The ZMP is the floor point where the horizontal components of the ground-wrench moment vanish. On this floor it coincides with the centre of pressure for a valid contact solution. Lift height and arm motion enter through centre-of-mass acceleration and angular-momentum rate.

Stability objective and constraints

2. A compact formulation suitable for an initial research proposal

4. Express ZMP in the support frame

Use a horizontal support frame S whose origin is the centre of the wheel footprint and whose axes follow chassis heading. Transform the world ZMP into that frame:

$$p_Z^S = R_2(\psi)^T (p_Z^W - o_S^W), \quad e_Z = p_Z^S - p_{Z, \text{ref}}^S$$

$R_2(\psi)$ is the planar rotation from S to the world horizontal frame; o_S^W is the footprint-centre position. The assumed four-wheel support region is $[-a, a] \times [-b, b]$. The quantities a and b come from contact geometry, not the visual chassis dimensions.

5. Proposed stability cost

$$\ell_{\text{stab}} = e_Z^T Q_Z e_Z, \quad Q_Z = \text{diag}(a^{-2}, b^{-2})$$

$$p_{Z, \text{ref}}^S = (0, 0)^T \implies \ell_{\text{stab}} = \left(\frac{x_Z^S}{a}\right)^2 + \left(\frac{y_Z^S}{b}\right)^2$$

$$J_{\text{stab}} = \frac{1}{T} \int_0^T \ell_{\text{stab}}(t) dt \simeq \frac{1}{N} \sum_{k=0}^{N-1} \ell_{\text{stab}, k}$$

Interpretation. The cost is zero at the support centre and increases as the ZMP moves away. Q_Z normalizes each direction by the available footprint. No independent tuning weights or additional tilt penalties are required in this initial formulation.

6. Feasibility and task requirements

$$|x_Z^S| \leq a - \delta, \quad |y_Z^S| \leq b - \delta, \quad 0 < \delta < \min(a, b)$$

$$\min_{u_{\text{cmd}}(\cdot)} J_{\text{stab}} \quad \text{subject to dynamics, task, contact, and actuator constraints}$$

The reserve δ accommodates estimation and model error. The cost encourages central support; the constraints preserve the reserve. Task constraints specify the desired motion and completion requirement; otherwise, remaining stationary can minimize stability cost.

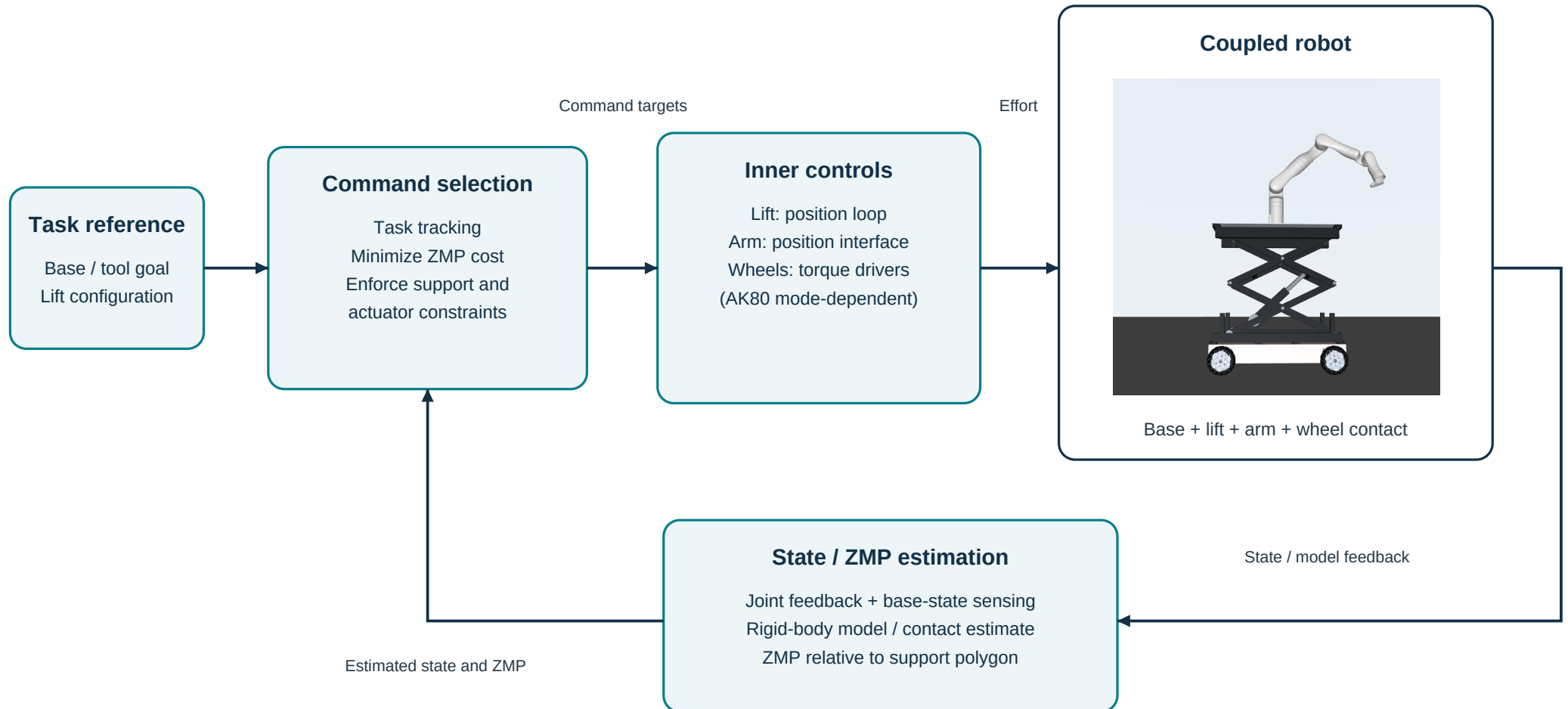
Additional constraints are the contact/friction law, positive supporting force, collision avoidance, and the actual joint, speed, torque, and lift-travel limits. If a wheel loses contact, replace the rectangle with the active contact polygon. ZMP containment alone is not a proof of recovery from arbitrary tilt or a guarantee against slip.

Notation for the cost

Symbol	Definition	Units
x_Z^S, y_Z^S	ZMP coordinates in the support frame	m
a, b	Support half-length and half-width	m
Q_Z	$\text{diag}(1/a^2, 1/b^2)$	$1/\text{m}^2$
δ	Reserved distance from each support edge	m
$T; N; k$	Motion duration; equal-interval sample count; sample index	s; -; -

Proposed hardware and feedback architecture

3. The robot image is the plant; controller choice remains open



Hardware command: $u_{cmd} = (a_{ref}, \theta_{ref}, T_{W,ref})$: actuator-extension target, arm-position targets, and wheel-torque targets. The lift potentiometer closes its position loop; it does not measure force. Torque-controlled wheels assume a compatible AK80 driver mode. Inner controls produce the mechanical force/torque inputs.